



Universiteit
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Bachelor Computer Science & Datascience and Artificial Intelligence

Ecosystem Simulation for Environmental
Policies using Reinforcement Learning

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RESEARCH PROPOSAL BACHELOR THESIS

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1 Background and theory

Environmental policy creation is crucial nowadays, as so many species are facing extinction and ecosystems are being damaged by improper practices and a lack of effective regulations. Such policy creation has traditionally relied on mechanical models, a set of ordinary differential equations. Lotka–Volterra, logistic growth, and system dynamics are among the most popular, and although easy to use and interpretable, they are limited. Thus, complex factors such as stochastic dynamics, nonlinear interactions, and adaptive behavior, extremely common in real-world ecosystems, are difficult, if not impossible, to model [TZC25] [Car17].

In recent years, reinforcement learning (RL) methods have appeared as promising alternatives for simulating and controlling complex ecological systems. RL allows agents to learn a policy based on the environmental feedback, closely resembling real behavior, and even allowing for the discovery of adaptive strategies on stochastic environments. Furthermore, RL allows for multi-agent environments, in which different entities can interact, making it a perfect choice for modeling ecosystems in which stability depends on the relation between the coexisting species [WCW20] [PLK⁺22].

Several recent studies have shown the viability of RL in ecological modeling. Wang et al. (2019) used deep RL to simulate co-evolution in a predator-prey system, showing how adaptive behavior contributes to ecosystem adaptability [WCW19]. Yamada et al. (2020) also explored large-scale multi-agent prey-predator environments and showed how RL frameworks can effectively simulate emergent ecological dynamics [YSTF20]. Finally, Park et al. (2022) demonstrated that proper environment definition and co-learning lead to realistic oscillatory population dynamics [PLK⁺22]. All this work shows that RL can adapt where classical models can not, simulating nonlinear interactions.

With the emergence of the intersection between RL and ecological modeling, this thesis aims to compare RL-based ecosystem simulations to traditional OED methods. Both in terms of accuracy and their potential to improve or inform environmental policy decisions [SGSA22]. For doing so, controlled environments with known species will be used, and benchmarked against replicable datasets and established biological modeling approaches.

2 Research question

Based on what was discussed in the previous section, we pose the following research question:

”How effective are reinforcement learning methods, compared to classical ecological models, in supporting ecosystem policy design?”

And state the succeeding sub-questions:

1. How accurately can Q-learning agents reproduce population dynamics in a predator-prey ecosystem compared to classical models?
2. How robust are the RL-derived population policies under noisy or perturbed environmental conditions, and how do they align with conservation objectives like species survival?

3 Study design/method

The study will implement and later compare two modeling approaches for simulating population dynamics. Primarily, with ordinary differential equations (ODEs) to simulate species interactions and population trajectories and estimations. These models will serve as the baseline.

The second approach will be applying tabular Q-learning agents in a grid-based environment, where species (the prey and predator) will interact, move freely, and take actions based on a partially observed environment. These agents will be trained using centralized training with decentralized execution (CTDE). Depending on the progress, this might be extended to use Deep Q-Networks (DQN) if time and requirements allow it.

The main environment will replicate known ecological systems using the empirical data available, and will recreate two case studies: the Isle Royale and the hare-linx system [MAR]. Both will have historical data for comparison and will provide a benchmark to validate the models.

The environment will be modeled as an MDP with the agent’s location, local population densities, and energy levels as states; moving, eating, and reproducing; and rewards will be tied to survival time, energy balance, and ecosystem stability. Evaluation metrics will include population stability, convergence speed, and comparison to ODE predictions.

The agents will be trained in randomly generated ecosystems based on their case study. Several reward functions and hyperparameters will be studied to yield the best and most accurate results.

4 Global planning

Date	Milestone / Task
6th March	Finalize the project proposal and design the first prototype of the environment.
22nd March	Have a working version of the main environment, analyze benchmark data, and review literature on dynamic models and environmental policy creation.
5th April	Implement the ODE models using the selected datasets and generate predictions.
19th April	Define the environment as a Markov Decision Process (MDP) and implement the tabular learning algorithm in the main simulation.
3rd May	Write the Introduction, Background, and Methodology sections of the thesis. Design multiple reward systems grounded in biological principles.
17th May	Apply the final reward system and train agents on noisy, procedurally generated environments. Run final simulations and generate graphical results.
31st May	Write the Experiments, Results, Conclusions, and Discussion sections. Submit draft for detailed feedback.
14th June	Use the received feedback to improve and refine the thesis document.
1st July	Finalize and complete the thesis with all necessary adjustments.
9th July	Present the Bachelor’s thesis at the project conference.

Table 1: Thesis Timeline

References

- [Car17] Bob Carpenter. Estimating lotka-volterra predator–prey dynamics with stan, 2017. October 16, 2017. <https://mc-stan.org/users/documentation/case-studies/lotka-volterra.html>.
- [MAR] MARSS package contributors. Isle Royale Wolf and Moose Data. <https://atsa-es.github.io/MARSS/reference/isleRoyal.html>. Accessed 2026-02-18.
- [PLK⁺22] Jeongho Park, Juwon Lee, Taehwan Kim, Inkyung Ahn, and Jooyoung Park. Co-evolution of predator-prey ecosystems by reinforcement learning agents. *Entropy*, 24(12):1682, 2022.
- [SGSA22] Daniele Silvestro, Stefano Gorla, Thomas Sterner, and Alexandre Antonelli. Improving biodiversity protection through artificial intelligence. *Nature Sustainability*, 5:415–424, 2022.
- [TZC25] Anna Treves, Giovanni Zenezini, and Elena Comino. Modelling system dynamics as a socio-ecological perspective to support human-beaver interactions, 2025.
- [WCW19] Xueting Wang, Jun Cheng, and Lei Wang. Deep-reinforcement learning-based co-evolution in a predator–prey system. *Entropy*, 21(8), 2019.
- [WCW20] Xueting Wang, Jun Cheng, and Lei Wang. A reinforcement learning-based predator-prey model. *Ecological Complexity*, 42:100815, 2020.
- [YSTF20] Jun Yamada, John Shawe-Taylor, and Zafeirios Fountas. Evolution of a complex predator-prey ecosystem on large-scale multi-agent deep reinforcement learning. *arXiv preprint arXiv:2002.03267*, 2020. <https://arxiv.org/abs/2002.03267>.